ML Model Evaluation & Explainability Report

Generated on 2026-10-18 10:57:34 | Session ID: cebad4d7-09d4-4b0d-aa6d-57a3c2ce05d5

1. Dataset & Problem Detection Summary

Dataset File:	house_prices.csv	**Total Rows:**	2000
Target Column:	Sale_Price	**Total Columns:**	9
Problem Type:	Regression	**Test Split:**	20%

2. Preprocessing & Validation Protocol

Numerical Imputer	Median	**Categorical Encoding**	Onehot
Categorical Imputer	Most_frequent	**Feature Scaling**	Standard
Random State	42	**CV Protocol**	5 Folds

3. Model Performance Comparison

Rank	Model	MAE	MSE	RMSE	R^2	Adjusted R^2	Train Time (s)
1	Linear Regression	20678.54	643499045.11	25367.28	0.9815	0.9809	1.25
2	Ridge Regression	20720.47	645316444.97	25403.08	0.9814	0.9809	0.88
3	Decision Tree Regressor	35175.00	2050668350.00	45284.31	0.9410	0.9393	0.88
4	Random Forest Regressor	25732.74	1036394108.52	32193.08	0.9702	0.9693	0.29
5	Gradient Boosting Regressor	22567.80	780642583.89	27939.98	0.9775	0.9769	0.25
6	XGBoost Regressor	24641.18	943488297.80	30716.25	0.9728	0.9721	0.19
7	Linear Regression	20678.54	643499045.11	25367.28	0.9815	0.9809	0.01
8	Ridge Regression	20720.47	645316444.97	25403.08	0.9814	0.9809	0.01
9	Decision Tree Regressor	35175.00	2050668350.00	45284.31	0.9410	0.9393	0.02
10	Random Forest Regressor	25732.74	1036394108.52	32193.08	0.9702	0.9693	0.26
11	Gradient Boosting Regressor	22567.80	780642583.89	27939.98	0.9775	0.9769	0.17
12	XGBoost Regressor	24641.18	943488297.80	30716.25	0.9728	0.9721	0.17

4. Recommended Champion Model

Recommended Model: Linear Regression
Criterion: Highest R^2 (0.9815).
Rationale: Superior performance and fold consistency on held-out validation data.

5. Global Feature Importance (SHAP)

Feature	Mean Absolute SHAP Value	Relative Impact
Square_Feet	154639.4085	100.0%
Year_Built	16621.9066	10.7%
Neighborhood_Suburbs	16551.3861	10.7%
Bathrooms	12389.5588	8.0%
Garage_Cars	11456.7181	7.4%
Condition_Rating	10963.2384	7.1%
Neighborhood_Uptown	10471.0549	6.8%
Neighborhood_Downtown	10190.5793	6.6%